

Polymathy and Neurodivergent Cognition

The Polymathic Neurodivergent Profile (PNP), the Capability-Adjustment Fallacy, and a documented live case at the intersection of clinical assessment, access to justice, and AI alignment research

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3 July 2026 · OSF: [10.17605/OSF.IO/6C5XB](https://doi.org/10.17605/OSF.IO/6C5XB)

Working status and disclaimer

This is a conceptual, clinical-formulation, and policy working paper. It is not a medical diagnosis, not a substitute for clinical assessment or legal submissions, and not a request for preferential treatment. "Polymath" is used throughout as a descriptive label, not a diagnostic category: it has no standing in psychiatric nosology, disability law, or educational classification. The single operative claim is functional: capability in one domain does not remove support needs in another.

ABSTRACT

This paper proposes the **Polymathic Neurodivergent Profile (PNP)** as a descriptive, non-diagnostic framework for a rare but practically important pattern: unusually strong cross-domain learning, structural pattern transfer between fields, intense self-directed research, and high output in complex, high-interest domains, coexisting with clinically significant difficulty in routine administration, pacing, timekeeping, task-switching, and low-stimulation procedural work. The central proposition is the **Capability-Adjustment Fallacy**: the institutional inference that visible high performance is evidence that reasonable adjustments are unnecessary. That inference is unsafe, and is contradicted by existing clinical guidance. NICE CG142 already directs professionals to recognise the discrepancy between intellectual functioning and adaptive functioning, and modern strengths-based assessment treats the same neurodivergent trait as advantage or impairment depending on context.

Beyond naming the pattern, the paper does three further things. First, it presents a **documented live case**: the author's own profile, evidenced across three independent documentation streams (clinical assessment, a court record in which adjustment requests and high-quality self-represented output coexist, and a cryptographically timestamped research corpus spanning AI alignment, scaling theory, and mathematics). Second, it sets out a

priority and convergence analysis: the framework's central claim about neurodivergent cognition was independently paralleled by Hernández-Espinosa and colleagues in PNAS Nexus (14 April 2026), whose argument that cognitive diversity is a protective factor for AI alignment in turn served as the stated conceptual foundation for a formal undecidability result (Gumbau Mezquita, June 2026); the author's timestamped manuscripts precede both. Third, it states a **falsifiability standard:** pre-committed abandonment conditions, operational measures, and the incremental-validity test the construct must pass or be discarded.

The aim is neither to medicalise polymathy nor to use ability as immunity from ordinary process, but to prevent a recurrent error: the conversion of a person's strengths into evidence against their needs.

Keywords: autism spectrum condition; ADHD; twice-exceptionality; monotropism; executive function; reasonable adjustments; strengths-based assessment; institutional misrecognition; access to justice; AI alignment; cognitive diversity.

CONTRIBUTIONS OF THIS PAPER

1. **A named, falsifiable descriptive construct (PNP)** with six operational components, each individually grounded in existing peer-reviewed literature, and pre-committed conditions under which the construct should be abandoned.
2. **A named institutional error (the Capability-Adjustment Fallacy)** with its clinical refutation (NICE CG142; DSM-5-TR masking; domain-specific executive function) and its legal refutation (Equality Act 2010 ss.20, 149(6); CPR PD1A).
3. **A three-stream documented case** in which the same person's clinical record, court record, and timestamped research corpus jointly instantiate the profile and the fallacy, making the framework checkable rather than anecdotal.
4. **A convergence analysis** connecting the profile to independent results in AI alignment: cognitive diversity as protective factor (PNAS Nexus, 2026) and the structural insufficiency of external verification (arXiv:2606.28639), with a precise, timestamp-anchored account of what preceded what.
5. **A field-level implication:** cognitive monoculture as an alignment risk, linking the clinical literature to the research programme's Papers III, IV.d and X.

1. Introduction

The contemporary neurodiversity literature has exposed a long-standing weakness in diagnostic and institutional reasoning: a tendency to recognise *difficulty* without adequately recognising *form, pattern, and unevenness* (Manalili et al., 2023). Autism and adjacent neurodevelopmental profiles are still often described primarily through deficits, even as a growing body of work argues that assessment should also identify strengths, skills, and interests, because these shape both lived experience and support needs (Woods and Estes, 2023; Russell et al., 2019).

This matters most for a particular kind of adult: one whose profile is neither global impairment nor smooth giftedness, but *radical asymmetry*: advanced performance in selected domains alongside marked dysfunction in others. This paper names one version of that asymmetry the **Polymathic Neurodivergent Profile (PNP)** and advances a single governing principle:

Capability describes what can be done under favourable cognitive conditions. Disability describes where a person is substantially disadvantaged by the environment, the process, or the demand structure. Both can be true at once.

That sentence is the functional heart of the paper. It is also the proposition that institutions, courts, clinics, universities and employers must understand before they can interpret such a person fairly. The paper goes further than naming the pattern: it documents a live instance across three independent record types, and it situates the pattern inside a set of 2026 results in AI alignment that have made cognitive diversity a research question rather than a slogan.

One biographical fact orients everything that follows. The documented case of Section 9 did not begin as an AI-safety researcher: he began as a *creation theorist*. The December 2024 manuscripts are, before anything else, a theory of why created systems persist: recursion amplifies, and only embedded, co-scaling correction prevents amplification from becoming destruction. The AI-safety programme, the scaling-law papers, the stability criterion and the book all *derive* from that single creation-theoretic move, and the derivation chain is timestamped end to end (HRIH states the hypothesis; Papers I to XI operationalise it). This matters for the present paper because it is the cleanest possible exhibit of PNP component 2, recursive pattern transfer: one structural idea, extracted once, applied across theology, cosmology, error correction, biology and machine learning, on checkable dates. The profile did not produce scattered interests; it produced one theory with many testable faces.

2. What this paper does and does not claim

Credibility requires the boundary of the claim to be stated plainly and kept.

It claims: that there exists a recurring, describable pattern of cross-domain high ability combined with disability-linked impairment in routine procedural functioning; that this pattern is currently under-described by deficit-only and credential-centric models; that describing it improves the accuracy of clinical formulation and the fairness of institutional treatment; that the supporting components are individually grounded in existing literature; and that one fully documented instance exists whose records are open to qualified scrutiny.

It does not claim: that "polymathy" is or should become a diagnosis; that PNP is a validated clinical entity (it is a hypothesis); that every highly able neurodivergent person fits the profile; that the profile confers superiority rather than unevenness; that documenting it guarantees any legal outcome; that the author's cognition caused any external research result; or that artificial intelligence creates the underlying cognitive pattern. PNP is offered as a descriptive construct to be tested, refined, or refuted, not as a discovery to be announced.

3. Background I: the turn to strengths-based assessment

The strongest single foundation for this paper is that good autism reporting is no longer expected to describe deficits alone. NHS adult-assessment guidance states that a report should record whether the person is autistic, what the person is good at, what they may need help with, and whether other conditions were found. NICE CG142 frames adult assessment as a comprehensive process covering childhood and adult presentation, functioning, coexisting conditions, sensory sensitivities, and attention to detail (NICE, 2021).

Woods and Estes (2023) developed the Survey of Autistic Strengths, Skills, and Interests precisely because clinicians and autistic people experience current approaches as overly deficit-centred and insufficiently informative for support planning; the neurodiversity paradigm reframes assessment toward what autistic people *can* do, not only what they cannot (Ferreira, 2025). Empirically, Russell et al. (2019) found that the most frequently described autistic traits, the ability to hyperfocus, attention to detail, strong memory, and creativity, "could be experienced either as advantageous or disadvantageous dependent on moderating influences." Cope and Remington (2022) similarly catalogue cognitive advantages in focus, memory, and creativity reported by autistic adults in the workplace.

This is the descriptive logic PNP sits inside. The point is not that strengths cancel difficulties; it is that strengths and difficulties are frequently *intertwined*, and that both are necessary to understand how a person functions. Neuro-affirming reporting guidance makes the same case at the level of practice, recommending that reports avoid crude functioning labels and instead describe strengths, difficulties, thinking style, and concrete adjustments (NAIT, 2024; Wilson and Gunn, 2023).

4. Background II: monotropism, hyperfocus, and the interest-based mechanism

If the strengths literature establishes *that* unevenness occurs, monotropism offers a mechanism for *why*. Murray, Lesser and Lawson (2005) describe autistic cognition as the tendency for attention to be pulled strongly, cognitively and emotionally, into a narrow attention tunnel, producing intense engagement with a small number of interests and difficulty engaging with stimuli outside the tunnel, with transitions experienced as costly. A self-report measure, the Monotropism Questionnaire, found both autism and ADHD associated with higher monotropism, with combined-profile respondents scoring highest (Garau et al., 2023; at the time of writing this measure was a preprint and is cited as preliminary).

Closely related, Ashinoff and Abu-Akel (2021) review *hyperfocus*, complete absorption in a task to the point where a person appears to tune out everything else, noting it is under-researched and absent from diagnostic criteria despite being ubiquitous in autism and ADHD self-report. In ADHD clinical writing, the same phenomenon is framed as an interest-based rather than importance-based pattern of attentional regulation; this framing is clinically useful but rests largely on clinical observation, and is presented here as explanatory rather than as hard evidence.

Together these accounts predict exactly the signature PNP describes: sustained, high-output engagement with self-directed, high-interest, high-complexity material, and disproportionate difficulty with externally paced, low-stimulation, low-reward procedural tasks. The complex task is not necessarily harder; for a monotropic mind it may be more *regulating*.

5. Background III: twice-exceptionality and asynchronous development

The gifted-education literature independently describes the same shape. "Twice-exceptional" refers to individuals who are simultaneously gifted and neurodivergent, whose spiky cognitive profiles show peaks in the gifted range and valleys in the impaired range, and who exhibit asynchronous development: an internal mismatch among cognitive, emotional, and procedural domains (Reis, Baum and Burke, 2014; Foley-Nicpon et al., 2011). A recent systematic review notes that asynchrony intensifies precisely when high intelligence coexists with significant weakness.

Critically, twice-exceptionality is explicitly a profile, not a diagnosis: a direct precedent for the way PNP is framed here. The literature also names the social consequence PNP is concerned with: such individuals can be *too able to be seen as disabled, and too disabled to be recognised as able*, and are therefore frequently misread by systems that expect ability to be uniform. Much of this literature is

child-focused and US-based; its extension to UK adults is offered as a reasoned generalisation, not as direct adult evidence.

6. The Polymathic Neurodivergent Profile: definition

A Polymathic Neurodivergent Profile is a proposed descriptive construct characterised by six interacting features:

#	Component	Description	Evidence strength
1	Cross-domain depth	Not casual interest in many subjects, but repeated acquisition of serious competence across distinct fields	Observational; polymathy literature (Araki, 2018; Root-Bernstein, 2009)
2	Recursive pattern transfer	Extracting relational structure from one domain and using it to accelerate learning or reasoning in another; "recursive" in the ordinary structure-mapping sense (Gentner, 1983; Gick and Holyoak, 1983). Genuine cross-domain transfer is rare and effortful, which is part of what makes a transfer-rich profile distinctive	STRONG mechanism literature; profile-level evidence observational
3	Autonomous research intensity	Deep self-directed study driven by intrinsic interest, often outside formal pathways	Consistent with special-interest depth (Grove et al., 2018)
4	Hyperfocus and immersion	Periods of unusually sustained attention, high output, and rapid synthesis in high-stimulation domains	STRONG (Ashinoff and Abu-Akel, 2021; Russell et al., 2019)
5	Executive asymmetry	Disproportionate impairment in routine administration, timekeeping, correspondence, filing, scheduling, task-switching, and procedural housekeeping	STRONG (NICE CG142 IQ-adaptive gap; Hill, 2004; Demetriou et al., 2018)
6	Institutional misrecognition	The tendency of organisations to misread the profile because its strengths and impairments violate ordinary expectations of consistency	STRONG (underemployment, masking and burnout literature; Buckland, 2024; ONS, 2021)

The framework is descriptive, not taxonomic. It implies no new clinical subtype, no biomarker, and no diagnostic threshold. Its purpose is narrower: to improve formulation where standard labels capture only fragments of the picture. In some cases "autism" or "ADHD" may name the neurodevelopmental substrate while "PNP" names the strengths-based manifestation and the misunderstandings that follow from it.

The obvious question is whether this is twice-exceptionality (§5) under a new name. It is not, and the difference is what the construct has to earn its keep on. Twice-exceptionality pairs general giftedness with a co-occurring disability, is overwhelmingly a child-and-education construct, and is silent about what follows institutionally. PNP specifies three things 2e does not: that the ability is *cross-domain structural transfer* rather than high ability in a single area (components 1 and 2); that the population is adults navigating administrative, employment and legal systems rather than schoolchildren; and that the pattern generates a *named, correctable institutional error*, the Capability-Adjustment Fallacy of §7, whose refutation is the paper's practical payload. Whether that specificity survives measurement is exactly the incremental-validity test §12 commits the construct to passing or failing in public.

In one sentence: *advanced capacity in complex, self-directed domains may coexist with serious impairment in externally sequenced, low-stimulation, administratively rigid systems.*

6A. The mechanism, stated humbly: trained recursion, not innate gift

This section is stated in the author's own voice; the disclaimer that opens it binds every clause that follows. Section 4 described the pattern from the outside, as monotropism. What follows describes the same pattern from the inside, because how the mechanism feels to its subject is part of what the profile has to explain.

The author does not claim superior intelligence. What is claimed is a different attentional architecture: perseverative, recursive iteration on the same problem or subject-matter until the outcome-space is substantially exhausted. Section 4 describes this pattern from the outside as monotropism; this subsection describes it from the inside. Every scenario is considered; every third-, fourth- and fifth-order influence is traced; every question is asked, answered, and then re-asked from a different angle. The loop does not release its object until the object stops yielding. This is not brilliance. It is a wiring, and a costly one, whose costs are the executive-function impairments recorded in §5 and evidenced across the case in §9.

The pattern produces the second mechanism: expertise as trained, not innate. Recursive questioning applied over decades to many domains accumulates by ordinary compounding. Since generative AI became widely available in 2023, the loop has been massively accelerated: answers that once required months of library work now return in minutes. More questions asked, more answers returned, more domains partially mastered, deeper transfer between them, expertise reinforcing expertise. The author was not born with cross-domain competence. The competence was ingrained, trained, and built up over decades of the loop running, and the polymathy documented in §9 is the accumulated output of that loop, not a starting condition.

The convergence is the sharpest new point in this subsection. The field has now demonstrated, on its own quantitative evidence, that sequential, iterative, self-questioning reasoning outperforms parallel sampling on the same problems. Sharma and Chopra published the quantified result in November 2025 (S. Sharma and R. Chopra, "The Sequential Edge: Sequential Recursive Processing Outperforms Parallel Sampling in 95.6% of Configurations", [arXiv:2511.02309](https://arxiv.org/abs/2511.02309), 4 November 2025). The author's independent formalisation appears in Paper II (The ARC Principle: Experimental Validation of Super-Linear Error Suppression Through Sequential Recursive Processing, January 2026), and the directional thesis is priority-anchored to the 8 December 2024 self-emailed manuscript bundle (SHA-256 f0d1f38f).

The standing three-layer crediting discipline applies here without exception. Publication priority on the quantified 95.6% number belongs unreservedly to Sharma and Chopra. Directional priority on the sequential-over-parallel thesis belongs to the author on the 8 December 2024 timestamp. Independent formalisation is acknowledged in both directions, with no influence claimed in either; the author was unaware of the Sharma and Chopra paper when he formalised the January 2026 experiments and encountered it months afterwards (biographical attestation, offered as such).

Reasoning-token architectures with visible chain-of-thought at scale (DeepSeek's R1, released January 2025, was among the first prominent open-weights instances of the type; the phrasing is deliberately conservative because reasoning-model work has multiple lineages) are the field engineering into machines the iterate-question-iterate loop that a monotropic mind runs natively. From this convergence a compact hypothesis follows, rung-tagged accordingly. **HYPOTHESIS** The neurodivergent cognitive style documented in this paper may be the biological instance of the reasoning architecture the field has just proved superior. If the hypothesis holds, it would explain both the polymathy (compounding cross-domain training) and the priority record recorded in §10: the loop was running in a single mind for years before the machines had it.

The claim is the polymathic *method*, never the stature. The book from which this paper descends (Eastwood, *Infinite Architects: Intelligence, Recursion, and the Creation of Everything*, print edition 2 January 2026, ebook 6 January 2026, ISBN 978-1806056200) already stated this position: its Author's Note (pp. 43-63) sets out the biographical basis and closes with a passage that binds the register.

The ARC Principle, the Eden Protocol, the framework of caretaker doping: these are my contributions. I make no claim to have invented the components. Recursion is well understood in mathematics. The alignment problem has been articulated by minds far more credentialed than mine. Religious traditions have explored stewardship for millennia. What I offer is the synthesis: the recognition that these separate streams are describing the same river, and that seeing them as one changes how we must act.

The book's Introduction names Rumi, Teilhard de Chardin and Leibniz as the three individual polymaths whose independent architectures converge on the same account. They were chosen precisely because they were cross-domain synthesisers, not as any claim to their stature. Claiming the stature of Rumi, Teilhard or Leibniz would be absurd, and this paper does not; borrowing their method, which is what a research posture actually does, is a different act altogether. This subsection is the formal statement in the profile paper of the position the book had already taken in prose: the polymathic method is claimed; the stature is not.

ILLUSTRATIVE VIGNETTE (ANECDOTAL, NO EVIDENTIAL WEIGHT CLAIMED)

Offered here strictly for orientation, and clearly outside the paper's evidential apparatus. A linear-thinking professional (the author's wife, a dentist) once agreed to role-play a student of the cross-domain framework. Her first question was which subject the lecture belonged to. The question has no answer, and that is the point. Cross-domain work has no home discipline, which is precisely why single-domain credentialled specialists were unlikely to have produced it, and why the Capability-Adjustment Fallacy's institutional home, "which box does this person fit?", is the same question in another costume. Section 7 states the fallacy formally; this vignette is only the domestic version.

6B. Retrospective trait-pattern assessment in the published literature, and the systemising mechanism at population scale

This subsection is placed strictly beside §6A because it is often the next question a careful reader asks: is the profile PNP names ever found in the historical record of cross-domain synthesisers? Two safety rules govern everything that follows.

The first safety rule is that this paper does not itself apply PNP, or any diagnostic category, to any named historical figure. The analytical work of connecting historical evidence to modern categories is done by others in credentialled venues, and this paper cites their work rather than reproducing it. The second safety rule is linguistic: the operative verb here is "consistent with", not "was". PNP itself takes no support from any individual retrospective case; the paper's interest in this literature is the pattern that credentialled retrospective analyses keep finding the same trait-cluster in history's cross-domain synthesisers. All crediting flows outward.

The retrospective literature is credentialled and it exists. Catani and Mazzarello (2019), a peer-reviewed analysis in *Brain* by clinicians at King's College London and the University of Pavia, examined the documented pattern in Leonardo da Vinci's records (chronic non-completion of

commissioned work, relentless procrastination and mind-wandering, productive bursts across day and night) and concluded that attention deficit hyperactivity disorder is the most scientifically plausible modern explanation of the recorded pattern (M. Catani and P. Mazzaello, "Leonardo da Vinci: a genius driven to distraction", *Brain*, 142(6), 1842-1846, [DOI 10.1093/brain/awz131](https://doi.org/10.1093/brain/awz131), 28 May 2019).

Two further monographs sit alongside it. Fitzgerald's *Autism and Creativity: Is There a Link between Autism in Men and Exceptional Ability?* (2004, Brunner-Routledge) surveys retrospective trait-pattern assessments for a broader sample of historical figures. James's *Asperger's Syndrome and High Achievement: Some Very Remarkable People* (2006, Jessica Kingsley Publishers) develops the same register for a set including Isaac Newton and Albert Einstein. Neither monograph diagnoses; each documents a recurring trait-pattern in the historical record and cites the biographical evidence that carries it.

Underneath the retrospective register sits a mechanism claim at population scale. Baron-Cohen's hypersystemising theory identifies the drive to detect if-and-then causal patterns as one of the engines of invention and cross-domain analytical pursuit (S. Baron-Cohen, *The Pattern Seekers: How Autism Drives Human Invention*, 2020, Basic Books, developed from a longer research programme on empathy and systemising in autism).

Genome-wide association work by Warrier, Baron-Cohen and colleagues reported a positive genetic correlation between autism and systemising of $r_g = 0.26$ (standard error 0.06, $p = 3.35 \times 10^{-5}$), independent of educational attainment and genetically dissociable from the social-communication component of autism (V. Warrier et al., "Social and non-social autism symptoms and trait domains are genetically dissociable", *Nature Communications*, 2019). The reported correlation does not diagnose any individual and does not settle prevalence for the profile this paper names. It does place the systemising drive characterising cross-domain analytical pursuit on a genetic axis linked to autism at the population level, rather than a purely biographical curiosity, and it is the kind of finding a reader can check.

The author's own working notes in the research corpus explored the mapping from DSM-5 ASC and ADHD criteria to the documented behaviours of celebrated historical polymaths, and contained their own section on the limits of retrospective methodology; those notes predate this paper but are cited here as working notes, not as evidence, and are consistent with the polymathic-method claim recorded in the book (§6A closing block).

Methodological caveat (stated prominently). Retrospective assessment of historical figures is contested in the clinical literature. The dead cannot be diagnosed. Historical

records carry period bias, translation loss, and selection: the records that survived may not be representative of the person's ordinary functioning, and the categories being applied are twenty-first-century constructs applied to lives lived under earlier ones. This paper accordingly treats the Catani-Mazzarello, Fitzgerald and James analyses as convergent observations by qualified authors, not as clinical facts, and PNP itself draws no support from any individual historical case. **HYPOTHESIS** The connection between the trait-cluster that credentialled retrospective analyses keep finding in history's cross-domain synthesisers, and the profile named in §6, is a hypothesis, not an established finding.

The closing tie is short. The book from which this paper descends (Eastwood, *Infinite Architects*, 2026a) chose Rumi, Teilhard de Chardin and Leibniz as its exemplars in the Introduction. The choice was principled, not decorative: those three were selected precisely because they were documented cross-domain synthesisers, and because the polymathic method (§6A closing block) is what they had in common. This subsection is that principle stated more carefully. The individual figures remain the property of the retrospective literature that examines them; the method is what is claimed, and it is claimed as method, not as stature.

7. The Capability-Adjustment Fallacy

The **Capability-Adjustment Fallacy** is the inference that because a person can perform a high-level task, they cannot require adjustments for lower-level procedural demands. It appears reasonable only on the assumption that cognitive difficulty scales linearly: that anything "easier" than the hard thing a person can do must also be within reach. In neurodivergent life, that assumption frequently fails. A high-complexity research problem may be *easier* than a simple administrative task, because the former is meaningful, patterned, self-directed, and intrinsically regulating, while the latter is fragmented, externally paced, low-reward, and executive-function heavy.

This is not a rhetorical move. It is anchored in clinical guidance. NICE CG142 directs all health and social care professionals to understand "the potential discrepancy between intellectual functioning as measured by IQ and adaptive functioning as reflected, for example, by difficulties in planning and performing activities of daily living including education or employment" (NICE, 2021). DSM-5-TR records that autistic difficulties "may be masked by learned strategies in later life," so that high apparent functioning can conceal, rather than refute, support needs (APA, 2022). The executive-function literature shows that differences in autism are real but domain-specific rather than globally uniform (Hill, 2004; Demetriou et al., 2018), and that everyday dysexecutive difficulties can be

present even where laboratory task performance is intact. A person can therefore be cognitively advanced and adaptively impaired at the same time; in PNP that discrepancy may be extreme and institutionally confusing.

The fallacy has a practical cost: it converts evidence of competence into evidence against need. A litigant who files a sophisticated written argument may be denied extra time, filing support, or procedural clarification because the decision-maker sees capability rather than disability. That is precisely backwards. The sophistication of the substantive work shows that the person's cognitive strengths are real; it says little about whether they can manage routine procedural burdens without disadvantage.

The proper test is **functional and demand-specific**: What is the task? What cognitive, sensory, social, administrative, and time demands does it impose? Which parts of the person's profile make that task disproportionately difficult? Which reasonable changes would reduce the disadvantage while preserving fairness to others and the integrity of the process? That is a more serious analysis than asking whether the person appears intelligent.

8. Institutional misrecognition: education, employment, courts, and AI

Education

A person may be in top sets for subjects that engage their interests and at the bottom in subjects that do not. Read morally, this becomes laziness or inconsistency; read neurodevelopmentally, it may reflect interest-based attentional regulation, executive asymmetry, and sensory or social load. Uneven attainment is common in neurodivergence, and strengths-based assessment argues that it should be described as a pattern rather than a character flaw.

Employment

The outcomes data are stark. Per the Office for National Statistics' first release to identify autistic respondents, autistic adults had the lowest employment rate of any disability group (ONS, 2021). The Buckland Review of Autism Employment found that only around three in ten working-age autistic people are in work, against roughly five in ten for all disabled people and eight in ten for non-disabled people; that autistic graduates are about twice as likely as non-disabled graduates to be unemployed fifteen months after graduating; and that autistic workers face the largest pay gap of any disability group (Buckland, 2024). Capability is plainly not being converted into recognition. Masking compounds this: camouflaging is associated with higher measured ability, later or missed diagnosis, and worse mental-health outcomes (Hull et al., 2017; Livingston et al., 2019), and

sustained mismatch between expectation and support is a route to autistic burnout: chronic exhaustion, skill loss, and reduced tolerance, distinct from ordinary occupational burnout (Raymaker et al., 2020).

Courts

Legal systems combine high cognitive complexity with intense procedural rigidity: deadlines, bundles, filing sequences, service evidence, transcript requests, order review. Those demands are executive-function intensive, not intellectually simple for everyone. A litigant in person can be strong in legal reasoning and weak in procedural administration; treating the former as proof that the latter needs no support produces unfairness, and is the Capability-Adjustment Fallacy applied to access to justice.

Artificial intelligence

Generative AI can accelerate drafting, research, and synthesis, but it does not originate the judgement, selection, cross-domain integration, and strategic direction that decide what to do with that output. For a person with a strong pre-existing pattern of cross-domain synthesis, AI functions as a *throughput multiplier*: it can make strengths more visible while leaving executive-function impairments untouched. The presence of AI-assisted work should therefore not be used to deny either the person's own cognitive contribution or their disability-linked support needs. This is the paper's most modest and least evidenced claim, and is offered as a reasoned position rather than a demonstrated finding.

9. A documented live case: three independent evidence streams

Named cognitive profiles usually rest on aggregated questionnaires or anonymised vignettes. This paper can do something unusual: present a single, fully documented instance in which the profile and the fallacy are recorded in three *independent* documentation streams, generated by different institutions for different purposes, none of them constructed for this paper. The subject is the author. Section 15 addresses the ethics and the epistemics of that choice; this section states the evidence.

Stream 1: the clinical record

The author holds adulthood diagnoses of autism spectrum condition and ADHD, with independent psychiatric documentation describing precisely the asymmetry the profile predicts: unusually strong self-directed learning, pattern recognition and cross-domain synthesis in high-interest domains, coexisting with clinically significant executive-function impairment in routine administration,

timekeeping and procedural work. The clinical documentation is held privately for obvious reasons, and is available to qualified reviewers under appropriate conditions. Nothing in this paper asks the reader to take the clinical stream on trust: the paper's argument survives even if the reader brackets it entirely, because the other two streams are public.

Stream 2: the court record

The author has conducted complex multi-track civil litigation in person in the courts of England and Wales. The record of that litigation contains, simultaneously: substantive written work of a standard the opposing parties engaged specialist counsel to answer, and repeated applications for modest procedural adjustments under the Equality Act 2010 and CPR PD1A, disclosed disability, and documented difficulty with exactly the procedural demand classes the profile predicts (deadlines, sequencing, administrative correspondence). The record therefore contains both halves of the fallacy in a single institutional file: output treated as evidence of capability, and adjustment requests assessed against that appearance of capability. The paper takes no position here on the merits of any live proceedings; the record is cited only as a documented instance of the demand-specific asymmetry, and it is a matter of court record rather than self-report.

Stream 3: the timestamped research corpus

The third stream is the one no questionnaire could fabricate: a cryptographically timestamped body of cross-domain research output. On 8 December 2024, the author self-emailed manuscripts totalling roughly 189,000 words; the export carries a published SHA-256 hash and a Google server timestamp via its Gmail Message-ID. The corpus that followed spans AI alignment (an evaluation-blinding protocol in which unblinding reversed the sign of a result), scaling theory (a co-scaling stability condition with a first real-model demonstration), mathematics (a functional-equation account of why two independent derivations of the metabolic scaling exponent converge, sent to the leading authors of both derivations), and physics-adjacent classification work on time crystals. Nineteen independently sourced external events have since converged on the corpus's structural claims; the register, with one primary source per row and priority explicitly credited to others where the dates demand it, is public.

WHY THE THIRD STREAM MATTERS METHODOLOGICALLY

Component 1 of the profile (cross-domain depth) and component 2 (pattern transfer) are ordinarily the hardest to evidence: self-report is unreliable and credentials measure institutional access, not cognition. A hash-timestamped corpus solves both problems. The dates cannot be backfilled, the domains are objectively distinct, and the transfer is inspectable in the documents themselves: the same recursion-and-correction structure is applied to quantum error correction, metabolic scaling, and reinforcement-

learning drift, years before independent groups published parallel structures in each domain. Whatever one thinks of the corpus's conclusions, its existence, dates and breadth are checkable facts, and they are exactly the facts the profile predicts and the credential system failed to register.

The three streams are independent in the relevant sense: a psychiatrist, a court, and a mail server have no common interest and no coordination. Their agreement on the same asymmetry, high structured output here, impaired procedural functioning there, is the strongest form of documentation available for a single case, and it converts the profile from an introspective claim into an auditable one.

10. Priority and the convergence chain

The framework's position in time can be recorded precisely, because 2026 produced independent results that bear directly on its central claim.

Date	Event	Anchor
8 Dec 2024	Author's manuscripts state the control-failure thesis for AI alignment and the embedded-correction remedy; the corpus that instantiates PNP components 1 to 4 begins its public timestamp record	SHA-256 published; Google server timestamp
May 2025	Hernández-Espinosa, Abrahão, Witkowski and Zenil circulate their preprint arguing that full alignment through uniform compliance is impossible on Gödel-Turing grounds and that cognitive diversity, explicitly including neurodivergent cognition, is a protective factor	arXiv:2505.02581
14 Apr 2026	The same argument is published in PNAS Nexus	DOI 10.1093/pnasnexus/pgag076
26 Jun 2026	Gumbau Mezquita publishes formal results (an unverifiability theorem and a soundness-completeness-tractability trilemma) whose stated conceptual foundation is the Hernández-Espinosa paper	arXiv:2606.28639
3 Jul 2026	This paper names the profile (PNP) and the Capability-Adjustment Fallacy, and places the documented case on the public record	This paper; OSF

Three points of precision. First, the author's manuscripts precede the preprint by roughly five months and the formal proof by eighteen; the direction of every gap is checkable against published hashes. Second, no causal claim is made in either direction: neither group had any way to know of

the author's work, and that mutual ignorance is what gives the convergence its evidential value. Third, the chain has a striking internal structure: a paper arguing that neurodivergent cognition protects alignment research from shared blind spots served as the stated foundation for the theorem that formalised what a neurodivergent researcher had argued structurally eighteen months earlier. The paper about minds like the author's inspired the proof that vindicated the author's thesis. The chain is documented end to end; every link is independently checkable.

11. Cognitive monoculture as an alignment risk

The convergence chain points at a field-level implication that goes beyond any single person. If Hernández-Espinosa and colleagues are right that uniform compliance cannot deliver alignment and that behavioural diversity is protective, then the composition of the research field itself becomes a safety variable. Uniform minds, trained in the same institutions on the same curricula, tend to miss the same things in the same direction at the same time. That is a monoculture risk, and it is invisible from inside the monoculture.

The research programme this paper belongs to offers three concrete instances. Paper IV.d showed that unblinded AI evaluation can reverse the sign of an alignment result, a failure mode that persisted across the field partly because evaluators shared assumptions about their own neutrality. Paper III mapped a single recursive-amplification structure across quantum error correction, biological scaling and machine reasoning, a mapping that required treating those literatures as one subject, which disciplinary training actively discourages. Paper X derived a stability condition for recursive self-improvement (correction must out-scale drift) whose central move, refusing to treat capability and correction as separable, cuts against the field's default decomposition. None of this demonstrates that neurodivergent cognition is necessary for such work. It demonstrates that the work in fact came from outside the monoculture, on documented dates, and that the field's own 2026 results now predict exactly that pattern.

The implication for institutions is uncomfortable and practical: exclusion of atypical cognitive profiles from research, funding and legal participation is not only an equity failure. On the field's own emerging evidence, it is a technical risk. The Capability-Adjustment Fallacy, operating in hiring committees, grant panels and courtrooms, is one of the mechanisms by which that exclusion happens.

12. Operationalisation and falsifiability

For the construct to be scientifically serious rather than self-descriptive, it must be operationalised and exposed to refutation.

Measurable components. PNP would be indicated by: (a) documented cross-domain output over time; (b) evidence of self-directed learning; (c) demonstrable pattern transfer between fields; (d) a measurable asymmetry between performance on preferred, high-interest versus non-preferred, low-stimulation tasks; (e) clinical evidence of executive-function impairment in routine administration; and (f) evidence of institutional under-recognition, where outputs and impairments are treated as mutually inconsistent.

Conditions under which PNP should be abandoned. The hypothesis fails if it cannot be reliably distinguished from (i) ordinary high openness to experience, (ii) ordinary giftedness, (iii) narcissistic or grandiose self-description, or (iv) general executive dysfunction; or if naming it adds no predictive or practical value beyond existing autism and ADHD formulations. These are real tests, and the paper welcomes them. The decisive empirical question is whether PNP *incrementally* predicts the preferred versus non-preferred task asymmetry and the institutional under-recognition outcome beyond what existing instruments already capture. If a validated measure showed PNP variance collapsing into those existing constructs, the construct should be discarded, and this paper's own falsification register commits to saying so publicly.

Research questions. Can clinicians and researchers reliably identify a cross-domain strengths-and-needs profile? How often does it appear among autistic and ADHD adults with histories of self-directed expertise? Does naming it improve clinical reporting, workplace support, educational accommodation, or court participation? The documented case in Section 9 supplies a template for the documentation standard such research would need: independent record streams rather than self-report alone.

13. Clinical relevance: what a report can legitimately say

The clinically defensible request is **not** that a psychiatrist diagnose "polymathy." It is that the psychiatrist provide a full adult autism assessment and, if clinically justified, describe the person's strengths, support needs, thinking style, executive-function profile, and functional consequences in real settings. That is consistent with NHS and NICE reporting principles and with neuro-affirming practice (NICE, 2021; NAIT, 2024).

A report could properly state, in substance, that the individual presents with autism-spectrum features and ADHD-related executive-function difficulties, together with an unusually strong profile of self-directed learning, pattern recognition, cross-domain synthesis, and productivity in selected high-interest domains; and that the *same* neurodevelopmental profile produces marked functional

disadvantage in routine administration, timekeeping, task-switching, and low-stimulation procedural work. The clinically important addition is the explicit statement that **the asymmetry itself is the finding**: high-level output should not be read as evidence that support needs are absent.

This formulation is more powerful than grand claims precisely because it remains sober, observable, and demand-specific. It asks the clinician to explain functioning, not to certify genius or legal correctness.

14. Legal relevance: adjustments restore equal footing, they do not confer advantage

The court-facing formulation should be precise and restrained, and should ask the court only to act consistently with its own access-to-justice obligations and with the functional evidence in any diagnostic report.

The statutory architecture supports this directly. The Equality Act 2010 imposes a duty to make reasonable adjustments where a provision, criterion, or practice places a disabled person at a substantial disadvantage (s.20), and in the exercise of public functions that duty is anticipatory. The Public Sector Equality Duty requires due regard to advancing equality of opportunity (s.149), and s.149(6) expressly provides that compliance "may involve treating some persons more favourably than others": statutory authority that adjustment is not unfair advantage. In civil litigation, CPR 1.1 and Practice Direction 1A require the court, so far as practicable, to ensure that parties are on an **equal footing**, can **participate fully**, and can give their **best evidence**, and recognise that vulnerability, including mental impairment or disability, may both impede participation and diminish the quality of evidence (CPR PD1A, 2021).

Within that architecture, the Capability-Adjustment Fallacy has a precise legal form: it substitutes a global impression of ability for the statutory question, which is always demand-specific disadvantage. A tribunal that reasons "this litigant writes able submissions, therefore no substantial disadvantage arises from these deadlines, this correspondence load, this hearing format" has answered a question the statute does not ask, and left unanswered the one it does. The correct sequence is the functional one set out in Section 7: identify the demand, identify the disadvantage on the evidence, identify the adjustment, and test it against fairness to the other parties. Nothing in that sequence turns on how clever the person's best work looks.

15. Limitations and ethics statement

Single-case evidence. Section 9 documents one instance, however thoroughly. A documented case can instantiate a construct and refute the claim that no such profile exists; it cannot establish prevalence, and it cannot by itself validate the construct. The paper's population-level claims rest on the cited literatures, not on the case.

Author as subject. The author is the documented case, which creates an obvious interest. Three mitigations are structural rather than rhetorical: the evidence streams are institutional records not authored for this paper; the falsifiability conditions in Section 12 are pre-committed and public; and the research programme this paper belongs to has already retracted one of its own quantitative claims when replication failed, with the retraction kept on the public record. Readers should nonetheless weight the case study as what it is: an unusually well-documented n of 1.

Privacy and dignity. Publishing one's own diagnostic status invites both stigma and its mirror image, inspiration narrative. The paper intends neither. The clinical stream is disclosed at the minimum level of detail the argument requires; the clinical documents themselves remain private. No inference should be drawn from this paper about any other person's profile, and the profile itself, as Section 2 states, confers no superiority: its costs are clinical, documented, and lifelong.

Live litigation. The court record is cited as an instance of a documented asymmetry, not as commentary on the merits of any proceedings, which remain for the courts that hear them.

What would change this paper. A validated instrument showing PNP variance fully absorbed by existing constructs; adult-population studies contradicting the asymmetry pattern; or failures of the documentation standard in Section 9 under qualified scrutiny. Any of these would be recorded on the programme's public falsification register.

16. Conclusion

The Polymathic Neurodivergent Profile is offered as a working hypothesis, not an established finding. It is falsifiable, it is open to correction, and it makes no claim to diagnostic status. It is a descriptive framework for a pattern that is real, institutionally consequential, and systematically misread; and it now has something such frameworks rarely have: a fully documented live case across independent clinical, legal and cryptographic records, and an independent 2026 literature, from formal logic to PNAS Nexus, converging on its central implication that cognitive diversity is not a concession but a resource.

The Capability-Adjustment Fallacy, the inference that visible high performance is evidence that adjustments are unnecessary, is unsafe. It is contradicted by clinical guidance, incompatible with the statutory architecture of the Equality Act 2010 and CPR PD1A, and, on the field's own emerging evidence, technically costly to the one research problem that can least afford shared blind spots. The

correction is not special treatment. It is the recognition that capability in one domain does not remove support needs in another, and that the same cognitive architecture which produces extraordinary output in favoured domains may be unable to produce consistent output in routine ones without structural accommodation.

The researcher who wrote this paper is the case study. The manuscript hashes are published. The court record exists. The clinical documentation exists. The convergence register is public, with priority credited to others wherever the dates demand it. The question this paper leaves with its readers is the one the fallacy prevents institutions from asking: not "how capable does this person seem?" but "what does this specific demand cost this specific mind, and what would fairness actually require?"

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ARC/Eden Research Programme · Paper C · Working Paper · 3 July 2026

OSF: [10.17605/OSF.IO/6C5XB](https://doi.org/10.17605/OSF.IO/6C5XB) · Evidence and falsification register: michaeldariuseastwood.com/evidence

Capability does not cancel disability. The asymmetry itself is the finding.